

The Citation Expansion Trap: Operating-Point Control for Fixed-Answer Multi-Hop RAG Citation Repair

Jiawei Li

Nanyang Technological University, Singapore
w250001@e.ntu.edu.sg

Abstract

Multi-hop retrieval-augmented generation systems often produce answers with incomplete or noisy citation sets. When the answer text is fixed, citation repair may seem to require only adding retrieved evidence. We show that this view is incomplete: in multi-hop settings, expansion that recovers missing support can also introduce plausible distractors into the evidence bundle. We study fixed-answer citation-set repair as a diagnostic protocol for reporting policies as operating points between support-ID coverage and distractor control. Across a calibrated 5,000-example HotpotQA diagnostic and Qwen2.5-generated outputs on HotpotQA, MuSiQue, and 2Wiki, cross-encoder repair yields cleaner support-ID operating points than BM25 expansion on HotpotQA and 2Wiki, while MuSiQue shifts the policy into a higher-distractor-cost regime under harder compositional transfer. We do not claim universal transfer of absolute performance from HotpotQA alone; the cross-dataset pattern is instead consistent with mechanism-level transfer of the support-distractor operating-point geometry. A single-author audit of 450 unique generated-output cases plus 60 duplicate checks reinforces this caution: support-ID coverage and claim-level citation support can diverge. We therefore recommend an operating-point scorecard for citation repair and separate support-ID diagnostics from stronger claim-level faithfulness claims.

1 Introduction

Retrieval-augmented generation (RAG) systems combine generated answers with retrieved evidence (Lewis et al., 2020). Many RAG interfaces now expose citations, and citation-aware generation benchmarks have made citation quality a measurable output property (Gao et al., 2023b). A citation can make an answer more inspectable, but it does not guarantee that the cited evidence is sufficient. Correct answers can still have unfaithful attributions

(Wallat et al., 2024). This issue is especially visible in multi-hop question answering: one passage may support the surface answer string, while another passage is needed to justify the bridge from the question to that answer (Yang et al., 2018).

We study *fixed-answer citation-set repair*: cases where the answer text is held fixed, but the attached citation set does not contain all evidence needed to support the answer under the available candidate passages. This setting is narrower than general attribution evaluation (Rashkin et al., 2023) and different from selective RAG systems that decide whether to answer, abstain, or mark an answer as unsupported (Qiu et al., 2026). It is useful when answer rewriting is expensive or undesirable, but it creates a distinct repair problem: add missing support passages without adding misleading distractors.

Our study asks three questions:

1. Can one-citation baselines expose multi-hop evidence gaps?
2. How quickly do added citations trade support recovery for distractor pollution?
3. Can reranking, calibration, and selective repair expose useful operating points?

We instantiate this study on HotpotQA (Yang et al., 2018), MuSiQue (Trivedi et al., 2022), and 2WikiMultiHopQA (Ho et al., 2020). One controlled setting fixes a gold answer with one gold-support citation to isolate citation repair from answer-generation errors. We also evaluate Qwen2.5-7B and Qwen2.5-14B generated outputs whose initial citation sets are produced before repair. Oracle repair estimates candidate-set availability; BM25 tests sparse retrieval; a DeBERTa cross-encoder tests learned support; and an API-based LLM verifier gives a small semantic comparison.

The study makes four contributions:

- We formulate fixed-answer citation-set repair as a post-generation problem and provide a diagnostic protocol for scoring support recovery, distractor pollution, and selective repair coverage.
- We identify the citation expansion trap: adding evidence to recover missing support can also introduce plausible distractors into the repaired evidence bundle.
- We evaluate repair policies as support-distractor operating points across a calibrated 5,000-example HotpotQA diagnostic and Qwen2.5-generated outputs on HotpotQA, MuSiQue, and 2Wiki.
- We check the diagnostic interpretation with a single-author audit over 450 unique cases plus 60 duplicate checks, framed as author adjudication rather than a separate multi-annotator evaluation.

The automatic metrics use dataset support annotations, not full claim-level faithfulness judgments. We therefore treat the numbers as support-ID diagnostics and use the author-adjudicated audit to probe proxy validity and failure modes rather than as standalone annotation evidence.

Our strongest calibrated diagnostic is HotpotQA, so we do not use it alone to claim universal transfer of absolute performance. Instead, we ask whether dataset shift preserves the support-distractor operating-point geometry closely enough to make calibrated citation repair choices meaningful across settings.

2 Related Work

RAG attribution and citation-aware generation. RAG combines generated answers with retrieved evidence, but provenance remains a persistent failure point (Lewis et al., 2020). AIS, ALCE, LongCite, and RAGAS make attribution and citation quality measurable (Rashkin et al., 2023; Gao et al., 2023b; Zhang et al., 2025; Es et al., 2024). Our target is a post-generation action: deciding which additional passages, if any, should be attached to a fixed answer.

Post-hoc revision and repair. RARR retrieves attribution for arbitrary model outputs and edits unsupported content while preserving the original text as much as possible (Gao et al., 2023a). Cite-Fix is the closest citation-correction neighbor: it

motivates post-hoc citation correction as a practical need and improves RAG citation accuracy by cross-checking generated citations against retrieved articles (Maheshwari et al., 2025). FaithfulRepair uses a narrower action space: it only edits the citation set for fixed multi-hop QA answers. Under this restriction, post-hoc correction exposes a multi-hop failure mode: adding citations can improve apparent support while degrading the cleanliness of the evidence bundle.

Evidence sufficiency verification. Answer correctness and citation adequacy are separable: a RAG answer may be factually right while its attached sources still fail to justify the claim (Waller et al., 2024). Document-level attribution methods ask which retrieved sources influence an answer (Nematov et al., 2026). The closest selective-verification neighbor is SURE-RAG, which treats retrieval as distinct from verification, aggregates pair-level claim-evidence checks into set-level evidence sufficiency, and supports selective RAG decisions about whether to answer or withhold an insufficiently evidenced answer (Qiu et al., 2026). Our setting is the repair-side counterpart: rather than deciding whether to answer, we ask how the citation set should move once the answer is already fixed.

Multi-hop support annotations. HotpotQA, MuSiQue, and 2WikiMultiHopQA offer multi-hop support annotations for controlled evidence diagnostics (Yang et al., 2018; Trivedi et al., 2022; Ho et al., 2020). QDRAG decomposes questions to improve grounding and reduce over-citation during generation (Liu et al., 2026). Generation-time query decomposition and reranking can reduce over-citation before answer production; FaithfulRepair shows that the same pressure persists as a post-generation repair risk when only the citation set can be edited. Our pilot uses support identifiers as weak diagnostics, not claim-level citation faithfulness; this gap motivates our claim-level audit sample.

3 Task and Method

3.1 Problem Setup

For each question x , we assume a candidate passage set P_x , an initial cited set C_x^0 , and a fixed generated answer. The repair system may add citations but does not rewrite the answer text. A repair action selects an added set $A_x \subseteq P_x \setminus C_x^0$, which

induces the repaired citation set $C_x = C_x^0 \cup A_x$.

Let G_x be the dataset support set for example x . We track two support-ID diagnostics,

$$S_x = \mathbf{1}[G_x \subseteq C_x], \quad D_x = \mathbf{1}[(C_x \setminus G_x) \neq \emptyset],$$

where $S_x = 1$ means that all gold support identifiers are cited after repair and $D_x = 1$ means that at least one distractor passage is cited. We also track the action size $|A_x|$, since larger repairs can recover support while also expanding the opportunity to attach unnecessary citations.

A compact per-example utility is

$$u_x(A_x; \lambda, \gamma) = S_x - \lambda D_x - \gamma |A_x|,$$

and the policy-level objective is

$$J(\pi; \lambda, \gamma) = \mathbb{E}_x[u_x(A_x^\pi; \lambda, \gamma)].$$

Here A_x^π is the added set chosen by policy π , λ prices distractor exposure, and γ is a small parsimony cost for adding citations. This is a policy-selection view, not a claim that deployed policies optimize the objective exactly. It makes the intended operating point explicit: recover missing support, avoid distractors, and abstain when additional citations are not justified. **Scorecard.** We report each policy with action coverage, support recovery, distractor exposure, action size, U_λ , and claim-audit alignment when available; detailed policy sweeps and uncertainty variants remain in the supplementary materials.

3.2 Baselines

Constrained citation baseline. We prompt the generator to answer in one concise sentence and use at most one citation. This creates a controlled evidence-insufficiency setting for multi-hop questions.

Oracle repair. Oracle repair adds the HotpotQA gold support passages. It is not deployable; it estimates the upper bound when the missing support evidence is available.

Lexical repair. Lexical repair ranks candidate passages by token overlap with the question and answer, then adds the top-ranked passages. This baseline tests whether simple matching is enough.

BM25 repair. BM25 repair ranks candidate passages with a standard unsupervised retrieval score over the question-answer claim proxy and passage text. This baseline tests whether sparse retrieval can recover missing support without a trained verifier.

3.3 Learned Verifier Repair

We train a DeBERTa-v3-base cross-encoder on claim–passage pairs derived from HotpotQA support annotations. The claim proxy is the question concatenated with the generated answer. Positive pairs are gold support passages; negative pairs are distractor passages.

At inference time, the verifier scores each candidate passage. We evaluate thresholded top- k repair policies that add high-scoring passages not already cited. We treat this model as a learned verifier baseline, not as a claimed replacement for sparse retrieval; one goal of the study is to test whether training improves the support–distractor operating point beyond BM25.

4 Experimental Setup

Dataset. The controlled and calibrated diagnostics use HotpotQA examples with questions, answers, candidate passages, and supporting passage identifiers. Each repair instance starts from a shuffled candidate set so that gold support passages do not always appear first. The 100-example slice is used for controlled stress tests and LLM-verifier comparisons; the main calibrated diagnostic uses 5,000 examples. The generated-output benchmark extends the same repair evaluation to HotpotQA, MuSiQue, and 2WikiMultiHopQA slices.

Controlled single-citation setting. Because repair keeps answer text fixed, we use a controlled 100-example isolation setting: each example starts from the gold answer with one gold-support citation attached. This creates an evidence-insufficient citation set whenever the multi-hop answer requires more than one support passage, while avoiding answer-generation confounds. Oracle repair can cite all missing gold support passages without adding distractors, so the bottleneck is citation selection rather than candidate-set availability. The controlled table does not estimate real-world under-citation frequency; it measures whether repair policies recover missing support without indiscriminately adding distractors. We also keep the citation-constrained LLM pilot as a diagnostic robustness check on generated answers.

Verifier training. The first verifier probe uses 600 claim–passage pairs from 100 examples: 200 positive support pairs and 400 negative distractor pairs. The pair file has a question-level train/dev split with 80 training and 20 dev examples. We

train DeBERTa-v3-base for three epochs on an NVIDIA A40 GPU. The main controlled table draws from the same broad example pool, so it is a pilot comparison rather than a fully held-out benchmark. To reduce overfitting risk, answer text, candidate passages, and the repair objective are fixed before repair; policies only add citations; BM25 is included as a non-parametric baseline; and separate held-out diagnostics are reported below. A full multi-domain benchmark remains future work.

Held-out calibrated diagnostic. To reduce dependence on the 100-example development pool, we construct a 5,000-example calibrated diagnostic with question-level train, calibration, and test splits of 3,000, 1,000, and 1,000 examples. This produces 29,967 claim–passage verifier pairs. We evaluate both BM25 and a DeBERTa cross-encoder trained only on the training split. The calibration split fits the abstention threshold calibration; the test split is held out for the reported fixed and selective repair points.

LLM verifier reference point. We also score the 100-example controlled pair file with an OpenRouter-hosted Gemini Flash Lite verifier prompt. This provides a semantic reference point for the lightweight local policies. The prompt asks whether a passage supports the question-answer claim proxy using only the passage, and returns a support probability. We report it as an API-based reference point, not as a local deployable component.

Evaluation metrics. Using dataset support IDs, **All Support Cited** means all gold passages are cited; **Incomplete** means at least one is missing; and **Unsupported/Distractor** means any cited passage is non-gold. These are pilot diagnostics, not final faithfulness metrics. We also audit 450 generated-output cases plus 60 duplicates to check proxy validity.

5 Results

Figure 1 reports the held-out 5,000-example calibrated support-ID diagnostic as a coverage-versus-quality trade-off. BM25 fixed top-1 reaches 0.820 support with 0.180 distractor citation; fixed top-2 improves support to 0.919 but nearly saturates distractor citation. Cross fixed top-1 reaches 0.975 support with 0.025 distractor citation. At $\theta = 0.93$, the selective Cross policy repairs 81.1% of test

Policy	Unique N	Full	Partial	Full+Partial	None	Distr.	Ans.
Baseline	150	0.400	0.413	0.813	0.187	0.293	0.520
BM25 top-2	150	0.400	0.453	0.853	0.147	0.893	0.507
Cross top-2, $\tau = 0.40$	150	0.507	0.313	0.820	0.180	0.407	0.507

Table 1: Single-author claim-support audit over 450 unique generated-output cases. Full/Partial/None denote claim support; Distr. marks any irrelevant or distracting citation; Ans. marks answer correctness. Sixty duplicates are used only for consistency checking and do not enter the main rates; exact agreement was 55/60.

examples and reaches 0.989 support with 0.011 distractor citation on accepted repairs. The held-out result supports a narrow claim: trained reranking plus abstention reduces citation pollution in this support-ID diagnostic.

The controlled 100-example stress test is retained in the supplementary materials as a mechanism check. It shows the same pressure in a harder fixed-answer setting: lexical top-2 reaches 0.770 all-support coverage but saturates distractor citation at 1.000, while the cross-encoder top-1 probe at $\tau = 0.20$ reaches 0.710 support with 0.280 distractor citation.

Figure 2 reports absolute support and distractor levels for Qwen2.5-14B generated outputs. For compactness, it shows the conservative cross top-2 row at $\tau = 0.4$; the seed-stable $\tau = 0.2$ operating point is discussed in Section 6 and the supplementary materials. BM25 top-2 reaches 0.955 support with 0.996 distractor citation on HotpotQA and 0.824 support with 0.930 distractor citation on 2Wiki, while Cross reaches 0.967/0.194 and 0.856/0.152, respectively. MuSiQue remains the harder boundary: Cross recovers support to 0.728, but its distractor rate remains 0.500. The full Qwen2.5-7B/14B policy matrix, audit-packet summaries, and seed/threshold diagnostics are retained in the supplementary materials.

The targeted $N = 28$ LLM robustness slice is also retained in the supplementary materials. It is illustrative rather than representative, but it shows the same failure mode: lexical top-2 raises all-support citation from 0.071 to 0.607 while citing distractors in every example, whereas the cross-encoder keeps the same all-support rate at $\tau = 0.20$ with lower distractor citation (0.464).

Table 1 reports a single-author audit over 450 unique generated-output cases plus 60 duplicates. The audit is stratified across both Qwen2.5-7B and Qwen2.5-14B, while Figure 2 shows the Qwen2.5-14B support-ID slice for space. It supports the proxy-based interpretation but also motivates cau-

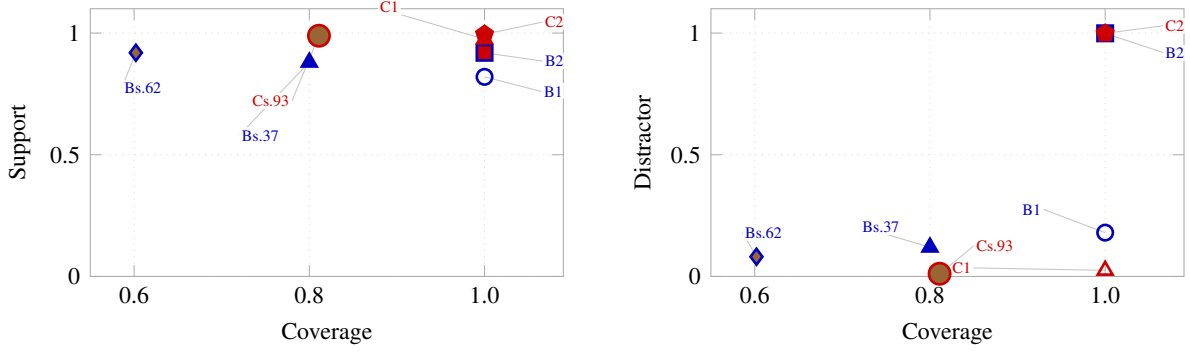


Figure 1: Held-out calibrated support-ID diagnostic. B1/B2 are BM25 fixed top-1/top-2; Bs.62/Bs.37 are BM25 selective thresholds; C1/C2 are Cross fixed top-1/top-2; Cs.93 is Cross selective repair. Selective rows report quality on accepted repairs, not unconditional full-coverage metrics.

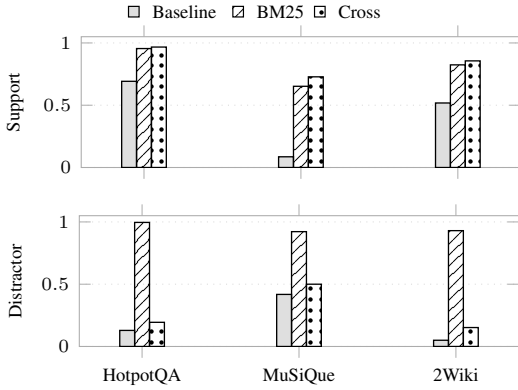


Figure 2: Generated-output absolute support-distractor levels on Qwen2.5-14B. BM25 buys support with distractor saturation, while Cross is cleaner on HotpotQA and 2Wiki; MuSiQue remains the higher-distractor boundary condition. Metrics are support-ID diagnostics.

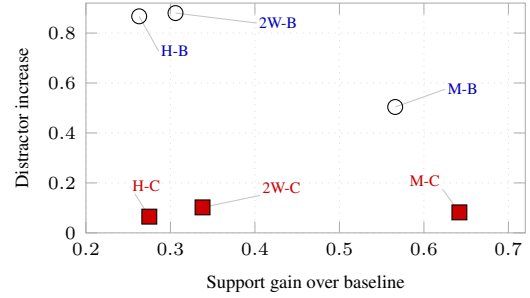


Figure 3: Generated-output support-distractor movement on Qwen2.5-14B. Moving right indicates support gain over baseline, and moving upward indicates distractor increase. H/M/2W denote HotpotQA, MuSiQue, and 2Wiki; B/C denote BM25 and Cross. BM25 usually buys support with larger distractor increases, while Cross occupies the cleaner operating point on HotpotQA and 2Wiki; MuSiQue remains the higher-distractor boundary condition.

tion: BM25 top-2 and cross top-2 have similar full-or-partial claim support (0.853 and 0.820), but BM25 top-2 cites distractors much more often (0.893 versus 0.407). Cross top-2 has the highest full-support rate among the audited policies (0.507), and baseline rows often receive partial support, reflecting missing bridge evidence. We use this audit to characterize support-ID proxy validity and failure modes, not as multi-annotator validation. A nonparametric bootstrap over the audited policy rows estimates the cross-BM25 distractor-rate difference at -0.487 $[-0.580, -0.400]$, while the full-or-partial support difference is -0.033 $[-0.120, 0.053]$; the audit therefore supports a cleaner operating-point interpretation. Duplicate rows are a consistency check, not evidence for the main rates: 55/60 received the same three author labels.

6 Analysis

The experiments separate two objectives that are easy to blur. Let $S_i(\pi)$ indicate that repair policy π cites all support identifiers for example i , and let $D_i(\pi)$ indicate that the repaired citation set contains at least one distractor. A compact operating-point view is $U_\lambda(\pi) = E[S_i(\pi) - \lambda D_i(\pi)]$, where λ is the cost assigned to exposing a reader to a distractor. The single-author audit gives a claim-support analogue of the same trade-off: BM25 top-2 has slightly higher full-or-partial claim support than cross top-2 (0.853 versus 0.820), but much higher distractor citation (0.893 versus 0.407), so the audit-side break-even cost is $\lambda^* = (0.853 - 0.820)/(0.893 - 0.407) \approx 0.068$. Thus the learned verifier is not a universal winner; it is the cleaner operating point whenever distractors carry even a modest cost.

Figure 3 complements the absolute levels in Figure 2 by showing movement from each baseline: moving right gains support, while moving upward increases distractor exposure. The label shorthand is defined in the caption. This view makes the operating-point shift explicit without restating the absolute rates.

Selective repair. The held-out calibrated diagnostic in Figure 1 provides the complementary selective-repair view. Fixed policies act at full coverage, while thresholds reduce action coverage and report support and distractor rates among accepted repairs. The point is not that abstention solves citation repair, but that citation-set repair should expose a controllable operating point rather than collapse support recovery and distractor exposure into one score.

The LLM verifier reference point shows that semantic judging can move the top-1 point closer to the oracle in the controlled setup. It does not remove the central trade-off: when forced to add two citations, the LLM verifier also saturates distractor citation. We therefore use it as an API-based diagnostic comparison point, not as the proposed lightweight repair component or as a replacement for claim-level human validation.

The BM25 result is also a warning for learned verifiers. In the controlled 100-example stress test retained in the supplementary materials, BM25 top-1 slightly outperforms the first cross-encoder probe at $\tau = 0.20$; the 5,000-example diagnostic reverses that pattern after a cleaner train/calibration/test split. We do not treat this as a contradiction. The controlled stress test is an entity hard-negative setting, where many distractors share entity names or topical words with the missing bridge evidence. Figure 1 asks a different question: whether a supervised scorer, trained and calibrated away from the test questions, can expose a cleaner operating point. In this diagnostic it can, but the lesson is the control mechanism rather than a universal model ranking.

Transfer boundary. The generated-output MuSiQue results sharpen this caution outside HotpotQA. On Qwen2.5-14B outputs, cross top-2 at $\tau = 0.2$ raises support from 0.086 to 0.774, but its distractor rate remains high at 0.524. This differs from the controlled MuSiQue smoke check retained in the supplementary materials, where the fixed-answer top-1 verifier transfers cleanly under support-ID diagnostics. The contrast suggests that the support-distractor tension is not only a Hot-

potQA artifact, while realistic out-of-domain generation remains harder.

Taken together, the strongest supported claim is not universal absolute-performance transfer, but mechanism-level transfer of the operating-point geometry. HotpotQA is our strongest calibrated support-ID diagnostic setting, so it gives the clearest positive case for operating-point control under fixed-answer repair. The generated-output results on 2Wiki and MuSiQue show that the same expansion pressure is not unique to HotpotQA: repair can still recover support under dataset shift, but the policy can move into a higher-distractor regime. In the utility view $U_\lambda(\pi) = E[S_i(\pi) - \lambda D_i(\pi)]$, this is exactly why Cross is not a universal winner in absolute terms, yet becomes the cleaner operating point once distractor cost exceeds a small break-even value. We therefore do not present FaithfulRepair as a general method for faithful generation; the contribution is to make support recovery, distractor cost, and calibrated operating-point choice visible and measurable for fixed-answer citation repair.

Robustness slice. The LLM-generated robustness slice makes the main failure mode concrete, but it is illustrative rather than representative. We do not use the 28 examples to estimate failure frequency; the 5,000-example calibrated diagnostic provides the quantitative evidence. Lexical top-2 often adds one support passage and one extra distractor, while the cross-encoder can recover the missing support more selectively or abstain at a higher threshold. The artifact examples show the same problem: surface overlap can find related passages, but it cannot decide whether the extra citation is necessary evidence for the answer.

Qualitative artifacts. The detailed qualitative table is retained in the supplementary materials rather than the archival main PDF. The recurring distractor types are still important: *surface-match distractors* share answer words or entity names but do not provide the missing bridge evidence; *entity-neighbor distractors* concern a related person, place, or organization rather than the needed relation; and *missing-bridge repair* adds a topically relevant citation that still leaves the comparison unsupported. Citation pollution is therefore not the same as answer correctness.

Proxy limits. Several examples also show a limitation of using dataset support identifiers as the only metric. Some generated answers make narrower claims than the full question, so a single citation can support the answer sentence even when

it does not cite all gold support passages. Conversely, over-citing all gold passages can still be stylistically or pragmatically undesirable. The author-corrected controlled sample and the 450-case generated-output audit check this issue directly; broader faithfulness claims would still need separate multi-annotator validation.

7 Limitations

Diagnostic scope. The controlled stress test uses a 100-example HotpotQA isolation setting. This isolates citation-set repair, but it is still a small diagnostic. The table includes examples from the same broad pool used for the first cross-encoder probe, so we treat those rows as feasibility evidence. The 5,000-example calibrated diagnostic adds a question-level held-out check for BM25 and the cross-encoder, but it is still one English multi-hop QA dataset rather than a multi-domain benchmark.

Transfer scope. Our strongest calibrated diagnostic is therefore HotpotQA, whose support annotations and candidate construction make it a favorable setting for controlled support-ID repair. We do not use HotpotQA alone to claim universal transfer. Instead, the relevant transfer object in this paper is the support–distractor operating-point geometry, not universal transfer of absolute scores. The Qwen2.5 generated-output benchmark over HotpotQA, MuSiQue, and 2Wiki suggests mechanism-level transfer of that geometry: repair can recover support beyond HotpotQA, but dataset shifts can move the policy into a substantially higher-distractor regime. MuSiQue is the clearest boundary condition for this effect rather than a counterexample to the main mechanism.

Proxy scope. The generated-output benchmark still uses support-ID diagnostics rather than full claim-level entailment, the LLM-generated constrained robustness slice is small at $N = 28$, and the single-author audit over 450 unique generated-output cases plus 60 duplicate checks should not be treated as full multi-annotator validation.

Verifier scope. The verifier is trained on question-answer claim proxies rather than manually segmented atomic claims. This keeps the pipeline lightweight but may miss fine-grained support failures. The support-ID proxy can fail in both directions: it can over-count support when a gold passage only partially supports the specific answer sentence, and it can under-count support when a generated answer makes a narrower claim than the

original multi-hop question. It can also treat stylistically unnecessary gold passages as required citations. The current repair policy only adds citations; it does not rewrite unsupported answers or abstain when the answer itself is wrong.

Language scope. All current results use English multi-hop QA; we do not claim generalization to other domains, retrieval settings, or citation styles.

8 Ethics and AI Assistance Disclosure

This work studies citation support, so the evidence trail matters. We keep result provenance in audit files and avoid treating unverified references or weak diagnostic metrics as final evidence.

We used AI-assisted tools for code drafting and debugging, experiment scripting, table preparation, language editing, consistency checks, and high-risk audit triage. The author selected the research direction, ran and inspected experiments, verified outputs, accepted or rejected audit suggestions, interpreted results, and is responsible for the final manuscript. No AI system is listed as an author.

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